

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Paper ID *****

Abstract

001 Long-horizon video generation is evaluated with whole-
002 frame metrics that reward motion and temporal consis-
003 tency. For fixed-camera nature scenes this creates an
004 ambiguity: motion of water, fire, smoke, or rain is de-
005 sirable, whereas motion of the background is an error.
006 A system can therefore score well on motion while its
007 scene drifts, or on consistency while its flow stagnates.
008 We introduce SNF-Bench, an evaluation framework for
009 long-horizon fixed-camera generation that partitions
010 each scene into static support and dynamic flow and
011 reports static fidelity, flow persistence with absolute
012 magnitude, and drift leakage separately and never as
013 one score, the last as an interpretive diagnostic rather
014 than a headline measurement. Each factor is validated
015 mechanistically rather than by correlation with prefer-
016 ence: we inject global translation, rotation, and scale
017 drift and progressive motion attenuation at known
018 severity into real generations, and require each factor to
019 respond in its stated direction and to remain selective
020 against corruptions it does not target. Auditing seven
021 publicly released long-horizon text-conditioned gener-
022 ators from their released checkpoints, plus an image-
023 conditioned deployment-sensitivity panel reported sep-
024 arately, we find that whole-frame motion and static-
025 region drift induce near-opposite orderings of the same
026 outputs, and that controlled drift which triples our
027 static-fidelity factors moves a widely used whole-frame
028 motion score by roughly a tenth—in the direction that
029 rewards it. SNF-Bench measures where motion occurs
030 and whether it persists; it does not measure physical
031 realism.

032 1. Introduction

033 Autoregressive and streaming video diffusion models
034 now extend generation from seconds to minute-scale
035 horizons. Evaluation has not kept pace: small tempo-

ral errors invisible in short clips accumulate over long
rollouts into background displacement, colour drift,
or motion decay, and whole-frame metrics summarise
these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature
scenes. In a static-camera video of a river, the wa-
ter should move while the bank, rocks, and surround-
ing landscape stay fixed. The same separation gov-
erns waterfalls, ocean waves, rainfall, fire, smoke, and
wind-driven vegetation: some regions form static sup-
port that must remain structurally fixed, while others
carry dynamic flow that must persist. Whole-frame
motion measures reward optical flow regardless of its
origin, crediting background drift as readily as genuine
fluid motion, while temporal-consistency measures can
favour videos whose stability comes from suppressing
the flow they should sustain. Two opposite failures—
static drift and flow decay—are thereby conflated, and
increasingly so at long horizons, where accumulated
background motion can dominate a sequence’s appar-
ent dynamics. Figure 1 shows both failures in released
generations of a single prompt.

We introduce SNF-Bench (Static-Natural Flow), an
evaluation framework built around this decomposi-
tion, evaluated at 60 and 120s horizons with shorter
and longer tiers retained as diagnostics. Measure-
ment is organised into three factor groups, the third
interpretive. Static fidelity captures geometric dis-
placement and residual flow inside nominally station-
ary regions. Flow persistence reports the retention of
dynamic-region motion with its absolute compensated
magnitude, so a ratio cannot look strong merely be-
cause motion is uniformly weak. Drift leakage relates
static-region motion and global-motion compensation
to dynamic-region flow; it contextualises the two mea-
surement axes rather than adding a third, and the cen-
tral claim does not rest on it.

A benchmark is only as trustworthy as its measure-
ments, so we validate the protocol independently of
any model comparison. Controlled perturbations in-

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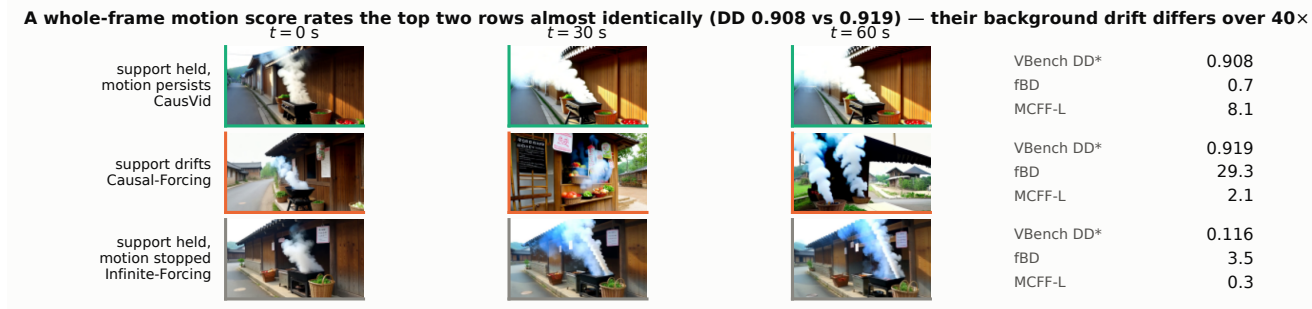


Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three released 60s generations of one prompt. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect. DD* is the per-video quantity underlying VBench Dynamic Degree, computed under its published rule but read before the binarisation the suite applies when aggregating, so that near-identical rows remain distinguishable as numbers.

ject global translation, rotation, and scale drift, progressive motion attenuation, late freezing, photometric drift, and temporal repetition into real fixed-camera generations at known severity, and we require each factor to respond in the direction its definition states and to remain selective against corruptions it does not target. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit seven publicly released long-horizon generators as executed through the recorded evaluation pipeline. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organised around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, never as a single score, the last as an interpretive diagnostic.

3. Mechanistic validation of every reported factor under corruptions of known type and severity, with selectivity measured against the families each factor should ignore and every measurement recomputed under erosion and dilation of the region partition.
4. A public-model audit showing that this decomposition materially changes the interpretation and ordering of long-horizon behaviour relative to whole-frame metrics.

2. Related Work

Video-generation evaluation. General-purpose suites assess visual quality, temporal consistency, motion, and semantic alignment. VBench [8] decomposes quality into dimensions including background consistency, motion smoothness and dynamic degree, and VBench-2.0 [26] extends it toward intrinsic faithfulness; EvalCrafter [13] learns a weighting of many metrics from user opinion; Video-Bench [4] scores generations with multimodal language models validated against human preference. Others specialise: T2V-CompBench [19] targets compositional binding, VMbench [11] perceptually aligned motion quality, and WorldScore [1] controllability and dynamics for world generation. These set the standards a benchmark paper must meet—explicit taxonomy, validated metrics, stated scale—but their measurements are whole-frame or sequence-level, and such an aggregate cannot reveal whether motion originates where intended or in displacement of nominally static structure. VMbench is the closest comparator, but it asks whether motion is perceptually plausible where we ask where it occurs and whether it survives compensation for global displacement. Our validation is correspondingly mechanistic—perturb by a known amount, require the expected response—complementing the human-alignment evidence those

147 suites provide.

148 Long-horizon autoregressive generation. Autoregres-
 149 sive and streaming generators extend synthesis beyond
 150 short bidirectional clips by producing frames or tem-
 151 poral chunks causally, including CausVid [24], Self-
 152 Forcing [7], Infinite-Forcing [9], Rolling-Forcing [12],
 153 Reward-Forcing [14], LongLive [22], and Causal-
 154 Forcing [27] with its few-step successor Causal-
 155 Forcing++ [25]; bidirectional systems such as LTX-
 156 Video [3] and Wan [21] provide short-horizon reference
 157 points. These systems differ substantially in temporal
 158 block size, training objective, memory and cache pol-
 159 icy, and horizon-extension mechanism [23], yet their
 160 evaluation typically relies on generic video metrics or
 161 method-specific qualitative analysis. That diversity
 162 motivates auditing each system in its released config-
 163 uration, and the evaluation gap motivates a system-
 164 agnostic question: does static support remain fixed
 165 while dynamic flow persists?

166 Image dynamics and nature-flow animation. Work
 167 on animating still images has long modelled natural
 168 motion explicitly: controllable fluid animation sepa-
 169 rates static image structure from specified fluid mo-
 170 tion [15], and Generative Image Dynamics models long-
 171 term scene motion through learned trajectory repre-
 172 sentations [10]. These works treat static support and
 173 dynamic content as distinct modelling targets; SNF-
 174 Bench elevates that distinction to an evaluation prin-
 175 ciple for long-horizon generators.

176 3. SNF-Bench Design

177 SNF-Bench evaluates long-horizon generation under a
 178 deliberately constrained setting: a fixed camera observ-
 179 ing a scene containing both stationary structure and
 180 persistent natural motion. This isolates two failure
 181 modes that are difficult to separate in unconstrained
 182 video—static drift, where nominally stationary struc-
 183 ture moves over time, and flow decay, where intended
 184 dynamics weaken or disappear. Every sequence is
 185 treated as a composition of two spatial roles, Ω_{static}
 186 for regions expected to remain geometrically station-
 187 ary and Ω_{flow} for regions expected to exhibit sustained
 188 motion. The partition defines an evaluation role, not a
 189 semantic ontology: a riverbank is static support while
 190 the river surface is flow; buildings are static support
 191 in a rainfall scene while the falling rain is flow. SNF-
 192 Bench does not require dynamics inside Ω_{flow} to follow
 193 any physical model. It measures whether motion stays
 194 localised to the appropriate regions and persists over
 195 time without contaminating static support.

Every system receives identical conditioning per
 item, and no prompt is rewritten for a particular
 method. Both conditioning modalities are evaluated
 under the same decomposition, differing only in how
 static support is established: the text-conditioned set-
 ting builds it in the opening of the sequence, while the
 image-conditioned setting takes it from a shared source
 image, so divergence there is attributable to the sys-
 tem rather than to a differently imagined scene. Both
 tracks are reported in the main text at the 60 and 120 s
 horizons; all four horizons appear in the supplementary
 material.

3.1. Scene Categories and Horizons

Prompts are stratified across representative fixed-
 camera natural processes—channel water, ocean waves,
 precipitation, fire and smoke, lava, and wind-borne
 motion—with the category set fixed before the final
 sweep. The realised composition is dominated by direc-
 tional and broad-area water motion with additional at-
 mospheric and combustion categories, and conclusions
 should be read at that scope. The strata are not equally
 populated: channel water accounts for 61% of the prin-
 cipal prompt set. Reported aggregates are means over
 prompts, not category macro-averages: three of the five
 populated categories hold two prompts each, so equal
 category weighting would buy balance at a cost in vari-
 ance the sample cannot support. The supplementary
 material instead recomputes every system both ways
 and states which conclusions survive, with complete
 per-category counts.

Four temporal regimes are used: 5 s as an initial-
 isation check, 60 s as the primary reported horizon,
 120 s as a supporting tier, and 240 s as a diagnostic
 extreme. Conclusions rest on the 60 s tier, which car-
 ries 23 prompts; the 120 s tier is reported alongside it
 as a supporting check, and the 5 and 240 s tiers—an in-
 itialisation check and a diagnostic extreme—are in the
 supplementary material. A system that cannot reach
 a horizon under its released configuration is reported
 as unsupported rather than silently modified.

3.2. Region Partition

The partition is obtained automatically and identically
 for every sequence. Mean optical-flow magnitude over
 the leading 12% of sampled frame pairs is thresholded
 by Otsu’s method [16] into a dynamic region and its
 complement; the static region is then eroded and a
 4% frame border discarded, which together form an ig-
 nored transition band so that boundary pixels—where
 the partition is least certain and flow least reliable—
 contribute to neither region.

Restricting the estimate to an early window prevents

247 circularity: deriving a partition from the entire motion
248 field would let accumulated drift be read as evidence
249 that the affected region is dynamic. At 60s the
250 partition is fixed from the first 7.1s and the metrics
251 evaluated over the remainder, so it reflects each system's
252 layout—unavoidable when every system authors
253 its own scene—but not the temporal failure being measured.
254

255 Two limitations bound the later conclusions. The
256 partition is automatic, so a system drifting within the
257 first 12% of its rollout can influence its own partition;
258 we therefore treat partitions as measured quantities
259 and recompute every result under systematic erosion
260 and dilation of the boundary (Sec. 5). And a two-way
261 partition cannot represent layered content: rain and
262 snow transit static support, so those pixels are assigned
263 by flow magnitude alone and static-fidelity values in
264 precipitation scenes carry an unmodelled contribution.
265 Derivation, the sensitivity study, and the annotation
266 protocol we regard as the natural strengthening of this
267 component are in the supplementary material.

268 3.3. Benchmark Manifest

269 SNF-Bench is defined by a versioned manifest fixed
270 before the final sweep, recording per item: identifier
271 and category, exact prompt, target horizons, resolution
272 and frame-rate metadata, and provenance. After
273 freezing, items are never removed because a system
274 performs poorly; corrections require a new benchmark
275 version. A failure taxonomy—unsupported horizon,
276 out-of-memory, runtime crash, numerical corruption,
277 invalid output—is likewise fixed in advance, and
278 failures are reported alongside successful generations
279 rather than silently excluded. Coverage is counted
280 from valid per-video measurements rather than from
281 the presence of a results file, so a run that terminated
282 early is reported as partial.

283 4. Drift–Motion Evaluation Protocol

284 SNF-Bench evaluates fixed-camera generation along
285 three factor groups: Static Fidelity and Flow Persistence
286 are the measurement axes, and Drift Leakage
287 (DLR, DAR) contextualises them rather than adding
288 a third measurement. No individual score is sufficient:
289 low background motion can be achieved by freezing the
290 entire video, and high foreground motion can be produced
291 by drifting the entire scene. We therefore always
292 report static-region stability jointly with both the relative
293 persistence and absolute magnitude of dynamic-region
294 motion.

295 Let I_t denote frame t , Ω_{static} and Ω_{flow} the region
296 masks of Sec. 3.2, and $\mathbf{u}_t$ the optical flow from I_t to
297 I_{t+1} , estimated with RAFT [20]. Each sequence defines

an early window $\mathcal{E}$ and a late window $\mathcal{L}$ of equal duration,
298 $\mathcal{E}$ beginning after a warm-up so that generation
299 transients do not define the motion reference; extents
300 are fixed in the specification.
301

302 4.1. Static Fidelity

303 Feature-aligned Background Drift (fBD). ORB features
304 [18] extracted inside Ω_{static} from the first frame
305 are matched against each late-window frame under
306 robust geometric filtering; fBD ($\downarrow$) is the median
307 matched-point displacement, averaged over the late
308 window and expressed as a percentage of the frame
309 diagonal. It captures accumulated geometric displacement
310 of persistent structure and is intentionally insensitive
311 to fine-scale flicker that moves no structure (formal
312 definition in Supp. S1).

313 Normalized Background Flow (NBF). Complementing
314 accumulated drift, NBF ($\downarrow$) is the mean optical-flow
315 magnitude inside the static mask, averaged over
316 all frame pairs and normalised by frame width for resolution
317 invariance and by the inter-frame interval Δt
318 for frame-rate invariance, giving units of 10^{-3} frame
319 widths per second (Supp. S1). Here Δt is the interval
320 of the sampled pairs actually used for flow estimation:
321 methods are evaluated at their native frame rates,
322 which differ across the roster, and the evaluation
323 pipeline subsamples each clip to a common temporal
324 rate before measurement. Expressing NBF per unit
325 time rather than per frame makes the quantity
326 independent of both the native rate and the sampling
327 policy, so the measure remains comparable if either
328 changes.

329 NBF is a normalised magnitude, not a ratio: dividing
330 by whole-frame flow would couple background leakage
331 to foreground behaviour and inflate the measure
332 for methods whose dynamic motion decays.

333 4.2. Flow Persistence

334 Global-motion compensation. Coherent scene displacement
335 is removed before dynamic-region motion is evaluated.
336 A single translation vector is insufficient: under rotation
337 or zoom the static-region flow field is not constant,
338 and its component-wise median is near zero for a centered
339 rotation, so a translation-only estimate would leave exactly
340 the corruptions Sec. 5 injects uncompensated. We therefore
341 estimate a robust global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$
342 from static-region correspondences under RANSAC, and
343 subtract the displacement field it induces from the measured
344 flow pointwise, giving a compensated field $\tilde{\mathbf{u}}_t$ (Supp.
345 S1). A similarity rather than an unrestricted affine model
346 is used because affine shear absorbs legitimate
347

348 local deformation and over-corrects. Where correspon-
349 dences are insufficient we fall back to median transla-
350 tion and record the fallback per sequence.

351 Motion-Compensated Foreground Flow (MCFF).
352 MCFF ($\uparrow$) is the mean compensated flow magnitude
353 inside Ω_{flow} over a temporal window. We report it on
354 both the early and the late window—MCFF-E and
355 MCFF-L—so a method with uniformly negligible motion
356 cannot appear strong through relative measures
357 alone (Supp. S1).

358 Flow Persistence (FP). FP ($\uparrow$) is the ratio of late to
359 early MCFF, clipped at a fixed constant to prevent
360 unstable ratios when early motion is small (Supp. S1).
361 FP near one indicates retained motion; substantially
362 smaller values indicate decay; values above one are not
363 automatically better. FP is never reported without its
364 MCFF magnitudes.

365 4.3. Drift Leakage and Attenuation

366 Whole-frame metrics cannot determine whether appar-
367 ent dynamic-region motion arises from intended flow or
368 from displacement of the whole scene; we report two di-
369 agnostics, since neither subsumes the other. Let F_{static}
370 denote mean flow magnitude over Ω_{static} , and F_{raw} ,
371 F_{comp} the mean raw and compensated magnitude over
372 Ω_{flow} , all in the late window, where drift has accumu-
373 lated.

374 Drift Leakage Ratio (DLR). DLR ($\downarrow$), the ratio of
375 static-region to raw dynamic-region flow magnitude in
376 the late window, locates the sequence’s motion energy
377 (Supp. S1). It is not a fraction and is deliberately
378 unbounded: $\text{DLR} > 1$ indicates that static support
379 moves more than the region intended to move, a regime
380 that occurs in practice.

381 Drift Attenuation Ratio (DAR). DAR ($\downarrow$), one minus
382 the ratio of compensated to raw dynamic-region flow,
383 measures the signed relative change in dynamic-region
384 motion after compensation—how much apparent motion
385 a global displacement model accounts for (Supp.
386 S1). Reported values are clipped to $[0, 1]$ while the
387 signed quantity is retained for validation; negative val-
388 ues, which arise where local flow opposes the estimated
389 global field, are reported rather than suppressed.

390 Interpretation. Neither is a causal decomposition of
391 motion into drift and non-drift components. DLR and
392 DAR are secondary diagnostics that divide the work

between them: DLR is a translational-leakage diag-
nostic whose rotational response is inverted, so rota-
tional drift is carried by fBD and DAR instead, and
that division is why DLR is never read alone. DAR
is signed, and its negative values are real. High DLR
with moderate DAR indicates static-region motion a
rigid model does not account for; it does not identify
what the residual is.

Semantic guardrail. The decomposition is agnostic to
what moves: a sequence rendering drifting fog where a
river was requested can occupy a favourable operating
point. The natural guardrail is a prompt-alignment
score—CLIP-based image-text similarity [6], or a
learned video-quality predictor [5] at higher cost. We
do not report one, and the factors should be read
accordingly: they establish where motion occurs and
whether it persists, not that the depicted phenomenon
is the one requested. Pairing SNF-Bench with a se-
mantic score is how it should be used in practice.

514 4.4. Joint Interpretation

No factor is read alone: low drift with low compensated
motion indicates an over-stabilised sequence, high com-
pensated motion with high static-region flow indicates
motion contaminated by displacement, and high reten-
tion with negligible magnitude indicates persistent but
trivial motion. SNF-Bench therefore reports no single
scalar score, since collapsing the factors would recreate
the ambiguity it exists to expose.

Implementation convention. Videos are normalised to
a common evaluation resolution and subsampled to a
common flow-estimation rate; no frame interpolation
is performed, since synthesising frames would alter the
flow field being measured. Every estimator, threshold,
window extent and constant is fixed globally before the
sweep and applied unchanged to every system, and the
per-sequence frame rate, resolution and compensation
mode are released alongside the scores.

520 5. Controlled Validation

Before SNF-Bench is used to compare generators, we
validate that its measurements respond to the failure
modes they target. A benchmark is not justified merely
because its metrics rank differently; each must change
selectively, monotonically, and in the expected direc-
tion under known corruptions. We therefore construct
controlled variants of fixed-camera reference clips with
known corruption type and severity, independently of
the audit in Sec. 6: perturbation strengths, masks, win-
dows, and hyperparameters are fixed before the final
results are inspected.

5.1. Perturbations and Criteria

Given a reference sequence we generate variants that isolate individual components of the decomposition. Geometric drift applies a temporally increasing global transformation—translation, rotation, or scale—simulating camera-like background displacement that accumulates as long-horizon error does. Motion attenuation progressively blends dynamic-region content toward an early reference, and late freezing replaces all content after a chosen fraction of the sequence, both leaving static support untouched. Photometric drift varies brightness and colour temperature while preserving geometry and motion, bounding the residual sensitivity of flow-based quantities. Temporal repetition replaces late dynamic content with a short repeated cycle. Partition perturbation erodes and dilates the region boundary rather than the video.

Geometric severity is specified as the mean pixel displacement a corruption induces, with each family’s parameter derived from it using the frame geometry. This is not presentational: a parameter in degrees is not commensurate with one in pixels, and at the evaluation resolution one degree of rotation induces only a few pixels of displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude.

A factor is admitted only if its response direction matches its definition, that response is monotonic in severity, unrelated corruptions produce substantially smaller changes, and conclusions survive perturbation of the region partition. Factors failing these checks are reported as diagnostics rather than headline quantities. The severity specification and the admission outcome for every factor are in the supplementary material.

Complementary checks. Mechanistic validation shows a factor responds to the corruption it names, not that its ordering matches what an observer perceives, and we do not close that gap. The supplementary material specifies an axis-specific protocol—forced choice on one failure axis, never on which output is better—and reports the one instance we ran: an open-weight judge [17] failed a presentation-order control, so no agreement rate is drawn from it. Perceptual agreement remains open.

5.2. Validation Results

Figure 2 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive late freezing drives MCFF-L and FP down—to 0.02 and 0.07 of their unperturbed values once the late window is frozen—while leaving the static factors comparatively

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	—

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25% on the factor’s target family. Rotational response is reported but not gated, and the factors divide that work explicitly: DLR is a translational-leakage diagnostic whose rotational response is inverted (−0.80), while rotational drift is carried by fBD and DAR at +0.70 and +1.00. That division is why DLR is never read alone. VBench DD is a comparison row rather than a candidate factor—it is put through the identical corruptions to show what a whole-frame score does under them—so its admission cell is left blank. Twelve reference clips per severity level.

unchanged, which is the selectivity the decomposition requires. The comparison that motivates the benchmark appears in the same experiment. Injected displacement that carries fBD to $3.2\times$ its unperturbed value moves VBench Dynamic Degree to only $1.1\times$, and in the wrong direction: a whole-frame motion score is largely insensitive to background drift, and to the extent it responds at all it improves on precisely the corruption a fixed-camera evaluation should penalise.

Two results qualify the framework rather than support it. Rotation is the harder geometric case for every factor, since correspondences degrade and the induced displacement is smallest at the frame centre: both static-fidelity factors respond more weakly than under translation (+0.70 for fBD, +0.60 for NBF, against +1.00 for both), so neither is used alone. DLR is the sharper limitation—admitted on its target family, where it responds perfectly, but with an inverted rotational response (−0.80), so it is read only as a comparative diagnostic alongside DAR. Reported alongside these is the scope boundary of Sec. 5.1: looping content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine progression from repetition, and does not claim to.

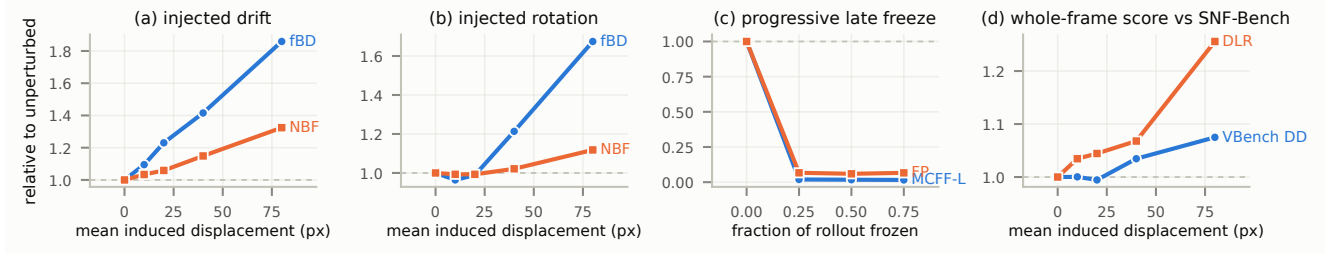


Figure 2. Mechanistic validation. Each panel injects a corruption of known kind and severity into real fixed-camera clips and plots each factor relative to its unperturbed value. Panel (c) shows the progressive late-freeze intervention the persistence factors are admitted on, so figure and criterion are one experiment: freezing the late window collapses MCFF-L and FP to a few percent of their unperturbed value. Panel (d) puts VBNch Dynamic Degree through the identical corruption as the SNF factors. Displacement that raises fBD to $3.2\times$ its unperturbed value moves the whole-frame score by $1.1\times$ —and upward: the generic score is close to blind to where the motion originates, and what little it registers it counts in the drifting system’s favour. The perturbation is synthetic, so the comparison implicates no published method.

6. Public-Model Audit

We apply SNF-Bench to publicly released long-horizon generators. The goal is not a universal leaderboard but a determination of how deployed systems occupy the static-fidelity/flow-persistence space under a single recorded deployment setting.

Systems and setting. The audit covers 7 publicly released autoregressive or streaming generators with accessible checkpoints and reproducible inference code, frozen as a set before the sweep. Unpublished and internal systems are excluded by construction, as are systems whose outputs cannot be independently reproduced. Every checkpoint runs through one common inference configuration rather than each method’s released defaults: a four-step denoising schedule, guidance 5.0, six frames per block and a fixed seed, with only the checkpoint and horizon varying. This is a matched-setting comparison and we describe it as one. It isolates the checkpoint, which is what makes the systems comparable, but a method distilled for a different step budget is not seen at its own operating point, so results are statements about these checkpoints under this configuration. The audited systems share a deployment envelope—all generate 832×480 at 16 fps and reach 240 s without modification to their released temporal policy—so no comparison here is confounded by output format. We never substitute a block size, cache policy or schedule to populate a table entry; a system that cannot reach a horizon unmodified is marked unsupported.

The image-conditioned track is audited under the same protocol and the same frozen specification. Systems evaluated under a common long-horizon wrapper rather than their released configuration are labelled

matched in every table, and their scores are read as deployment sensitivity rather than as the published method’s performance; we never rank a native row against a matched one.

Controls and statistics. Within each item every system receives identical conditioning, and the seed set is fixed before the sweep. Seed control guarantees reproducibility within a system, but because implementations consume randomness differently, identical seeds do not induce matched noise draws across systems; cross-system comparison therefore rests on the paired-item design. Videos are retained in native form and the normalisation of Sec. 4 applied unchanged.

Scores are computed per item and averaged over items; the sensitivity of every conclusion to category macro-averaging is reported in the supplementary material. Uncertainty is estimated by percentile bootstrap [2] over items, and we report 95% intervals. Exact ordinal ranks are avoided where uncertainty does not support them: systems whose intervals overlap substantially are not described as separated. Operational limits—unsupported horizons, failed generations, memory exhaustion, invalid outputs—are reported as rates against the frozen manifest and never removed after inspection.

To test whether the decomposition changes interpretation, the same outputs are additionally scored with established whole-frame metrics and the induced orderings compared. The focus is interpretation change, not a claim that existing metrics are wrong.

7. Results and Interpretation Changes

Unless stated otherwise, comparisons refer to systems under the recorded deployment setting; deployment-

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 23$)				
CausVid	8.08 ± 3.15	9.27 ± 4.14	2.87 → 1.96	0.67 ± 0.18
Self-Forcing	11.4 ± 3.63	12.7 ± 3.97	10.4 → 2.54	0.44 ± 0.22
Infinite-Forcing	7.75 ± 3.84	3.65 ± 0.88	1.17 → 0.29	0.63 ± 0.22
Rolling-Forcing	16.8 ± 3.69	11.2 ± 2.70	3.18 → 1.57	0.75 ± 0.29
Reward-Forcing	14.8 ± 3.71	8.51 ± 2.79	3.81 → 0.80	0.49 ± 0.18
LongLive	15.9 ± 2.89	13.9 ± 4.76	6.46 → 2.41	0.75 ± 0.28
Causal-Forcing	22.4 ± 2.26	87.6 ± 18.6	9.84 → 7.21	0.83 ± 0.31
120s horizon ($n = 6$)				
CausVid	6.15	3.42	0.36 → 0.52	1.06
Self-Forcing	21.2	6.80	4.85 → 0.82	0.85
Infinite-Forcing	16.1	2.60	0.50 → 0.32	0.64
Rolling-Forcing	23.8	10.6	3.33 → 0.77	0.38
Reward-Forcing	20.0	11.6	2.05 → 0.69	0.39
LongLive	22.5	9.66	5.10 → 1.19	0.81
Causal-Forcing	21.8	130	10.5 → 10.7	0.90

Table 2. SNF-Bench audit of public T2V checkpoints under the recorded deployment setting. All systems are public external models executed through one evaluation pipeline; the manifest executed under one common inference configuration—four-step schedule, guidance 5.0, six frames per block, fixed seed—rather than each method’s released defaults, so these are matched-setting results for these checkpoints and not a reconstruction of any author’s deployment. Unpublished systems are excluded by construction. DAR is reported clipped to $[0, 1]$; signed values are released. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel. Values are means over prompts $\pm$ half a percentile bootstrap 95% interval (10k resamples, fixed seed); panels with fewer than ten prompts report the point estimate alone, since an interval resampled from so few items is not informative. MCFF-E and MCFF-L are given together because FP is a ratio and is not interpretable without the magnitude it is taken over; intervals overlap for several systems, and only non-overlapping comparisons are described as separated. Aggregation sensitivity and the DLR/DAR diagnostics are reported in the supplementary material.

sensitivity results are reported separately in the supplementary material and never redefine a system’s native behaviour.

7.1. Native-Capability Audit

Table 2 reports the text-conditioned track at 60s, the tier whose sample supports method-level separation, and at 120s as a supporting tier. The 120s panel carries 6 prompts and is read as corroboration of direction rather than as an independent separation; it draws a different prompt set from the 60s panel, so the two panels are not compared column-by-column. The 5 and 240s tiers, an initialisation check and a diagnostic extreme respectively, are in the supplementary material with all four horizons.

Motion persistence. Dynamic-region behaviour is read at the joint point (MCFF($\mathcal{L}$), FP), never FP

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 30$)				
Causal-Forcing++	9.18	5.97	0.85 → 0.60	0.86
(2-step) [†]				
Causal-Forcing++	5.34	7.05	0.86 → 0.57	0.77
(1-step) [†]				
Causal-Forcing (frame-wise) [†]	21.1	36.4	2.80 → 2.69	0.89
Causal-Forcing++ (2-step, frame-wise)	9.94	6.91	1.30 → 0.78	1.02
Self-Forcing [†]	17.3	5.52	1.50 → 0.69	0.89
CausVid [†]	19.9	3.31	0.58 → 0.41	0.81
Causal-Forcing++ (1-step, frame-wise)	10.4	5.25	0.53 → 0.52	1.01
120s horizon ($n = 20$)				
Causal-Forcing++ (2-step) [†]	9.65	8.81	0.89 → 0.82	0.93
Causal-Forcing++ (1-step) [†]	7.26	9.39	0.84 → 0.66	0.86
Causal-Forcing (frame-wise) [†]	19.9	52.7	4.82 → 5.06	1.08
Causal-Forcing++ (2-step, frame-wise)	11.4	17.3	1.89 → 1.54	1.01
Self-Forcing [†]	23.0	10.3	2.41 → 0.76	0.45
CausVid [†]	19.7	6.04	1.15 → 0.83	0.67
Causal-Forcing++ (1-step, frame-wise)	13.6	7.39	1.01 → 0.35	0.64

Table 3. Image-conditioned track: native capability and deployment sensitivity. Unmarked rows are native-capability results; [†] rows are a deployment-sensitivity panel, not an audit of those systems. native systems run under their authors’ configuration, matched systems under the common long-horizon wrapper and are read as a deployment stress test rather than as the published method’s performance. [†] marks a system run under the common long-horizon wrapper rather than its released configuration; such rows are read as deployment sensitivity and are never ranked against native rows. n is stated per horizon and is common to every row of that panel. Values are means over prompts; intervals for this track are given with the complete per-horizon tables in the supplementary material. MCFF-E and MCFF-L are given together because FP is a ratio and is not interpretable without the magnitude it is taken over; intervals overlap for several systems, and only non-overlapping comparisons are described as separated. Aggregation sensitivity and the DLR/DAR diagnostics are reported in the supplementary material.

alone: sustained flow needs non-trivial late motion and high retention. MCFF-E beside MCFF-L exposes the degenerate case, since negligible motion throughout cannot reach a high FP without a small MCFF-E.

Operating regimes. Read jointly, the static-fidelity and persistence columns separate low-drift/low-motion, high-motion/high-drift and low-drift/non-trivial-motion behaviour without collapsing the axes into a scalar score; the supplementary material plots this. The extremes are informative in opposite directions. Infinite-Forcing has the lowest normalised background flow: a paired bootstrap over per-prompt dif

ferences separates it from all six other systems, every interval excluding zero (supplementary material). It also shows the least surviving late motion; on feature-based drift its interval overlaps the next-ranked system, so it is not the least-drifting system outright. Causal-Forcing sits alone at the opposite corner, separated on background flow by a wide margin.

7.2. Drift Leakage Changes the Interpretation of Apparent Motion

The effect is systematic rather than anecdotal. Correlating each generic whole-frame metric with each SNF-Bench factor over method-level means (supplementary material), Dynamic Degree tracks static-region flow at $\rho = +0.93$, so the systems a generic metric reads as most dynamic are, closely, those whose backgrounds move most. Causal-Forcing ranks 1 of 7 on Dynamic Degree and 7 on NBF; Infinite-Forcing mirrors it, ranking 1 on static fidelity and 7 on apparent motion. Read with its magnitudes, Causal-Forcing’s DLR of 0.94 indicates static-region motion energy almost equal to the region intended to move, while its DAR of 0.40 indicates that a rigid global model accounts for only part of that motion. What the residual contains is not identified here—it may include non-rigid instability, intended local motion, partition contamination or flow error—and these measurements do not separate them.

7.3. Image-Conditioned Track: Native Capability and Deployment Sensitivity

Table 3 is two things, kept apart. The two unmarked rows are a native-capability result: systems whose released configuration reaches these horizons, measured as shipped. The five marked rows are a deployment-sensitivity panel: released checkpoints whose own configuration does not reach these horizons, run under a common long-horizon wrapper. This is not a seven-system audit and is not presented as one; ranking a marked row against an unmarked one would compare one system’s design against another’s deployment. The panel supports one descriptive observation: the same checkpoint under a wrapper rather than its own released setting yields materially different scores, so a conclusion about a system is a conclusion about it as deployed. Why that happens is outside what a benchmark establishes.

The track supplies what the text-conditioned one cannot: every system starts from the same source image, so static support is given rather than invented and divergence is attributable to the system. Bidirectional reference systems reach only the 5s tier and are reported there.

8. Limitations and Conclusion

SNF-Bench is intentionally narrow: long-horizon fixed-camera generation, where static support and dynamic flow separate. It measures neither physical realism nor semantic correctness, cannot distinguish progression from repetition, and rests on an automatic partition whose limits are stated in Sec. 3.2 and bounded by the erosion-dilation study. Conclusions unstable under that perturbation are never headline claims.

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